

```
# taqman3c
```

```
0.4.0
```

```
3C / 3C-qPCR / TaqMan Anchor-Test
```

```
---
```

```
## 1.
```

```
'taqman3c' R 3C TaqMan qPCR
```

- Anchor +
- Test
-
- FASTA
- BEDPE 7 A/B Anchor

```
---
```

```
## 2.
```

```
### 2.1
```

- R >= 4.2
- 'devtools'
- 'testthat' DESCRIPTION Suggests

```
### 2.2
```

```
--- code ---
```

```
install.packages("devtools")
```

```
devtools::load_all(".")
```

```
--- code ---
```

```
### 2.3
```

```
--- code ---
```

```
install.packages("devtools")
```

```
devtools::install(".")
```

```
library(taqman3c)
```

```
--- code ---
```

```
### 2.4
```

```
--- code ---
```

```
devtools::test()
```

```
--- code ---
```

```
--- code ---
Rscript -e "devtools::test()"
--- code ---
```

3.

3.1 sequence

- DNA 'A/C/G/T'
-

3.2 API BEDPE

- 'anchor_region = c(start, end)' 1-based
- 'anchor_cut_site' anchor_region
- 'test_regions'
- 'region_id'
- 'start'
- 'end'
- 'cut_site'

3.3 BEDPE

BEDPE

1. chr1
2. start1
3. end1
4. chr2
5. start2
6. end2
7. anchor_flag

'anchor_flag'

- 'A' Anchor2 Test
- 'B' Anchor1 Test

> BEDPE start 0-based end 1-based/exclusive

```
--- code ---
chr1  5000  6000  chr1  10000  11000  A
--- code ---
```

```
## 4. API'design_3c_taqman_assays()
```

```
### 4.1
```

```
--- code ---
```

```
seq <- "ACGT..."
```

```
test_regions <- data.frame(  
  region_id = c("enh1", "enh2"),  
  start = c(800, 1200),  
  end = c(1100, 1500),  
  cut_site = c(980, 1360)  
)
```

```
assays <- design_3c_taqman_assays(  
  sequence = seq,  
  anchor_region = c(100, 400),  
  anchor_cut_site = 320,  
  test_regions = test_regions  
)
```

```
print_assays(assays)
```

```
--- code ---
```

```
### 4.2
```

```
- 'max_results'  
- 'primer_len_range' 'c(18, 30)'  
- 'probe_len_range' 'c(20, 30)'  
- 'primer_tm_range' Tm 'c(58, 62)'  
- 'probe_tm_delta' Tm 'c(8, 10)'  
- 'primer_gc_range' GC% 'c(30, 80)'  
- 'distance_to_cut' 'c(50, 150)'  
- 'probe_gap_range' anchor 'c(1, 10)'  
- 'amplicon_size_best' 'c(60, 150)'  
- 'amplicon_size_max' '300'  
- 'secondary_enzyme_sites' motif
```

```
---
```

```
## 5. BEDPE API'design_3c_from_bedpe()
```

```
### 5.1
```

```
--- code ---
```

```
assays <- design_3c_from_bedpe(  
  sequence = seq,  
  bedpe_path = "pairs.bedpe",  
  sequence_genomic_start = 1,  
  region_padding = 0  
)
```

```
--- code ---
```

5.2

- 'bedpe_path' BEDPE
- 'sequence_genomic_start' 'sequence' 11-based
- 'region_padding' BEDPE
- '...' 'design_3c_taqman_assays()'

5.3 A/B

- BEDPE
- 7 A/B
- 'sequence_genomic_start'

6.

6.1

--- code ---

```
assays <- design_3c_taqman_assays(  
  sequence = seq,  
  anchor_region = c(100, 400),  
  anchor_cut_site = 320,  
  test_regions = test_regions,  
  uniqueness_genome_fasta = "genome.fa",  
  uniqueness_max_hits = 1,  
  enforce_probe_uniqueness = FALSE  
)
```

--- code ---

6.2

- exact match
-
- 'uniqueness_max_hits=1'
-
-

7. CLI

7.1 BEDPE CLI

--- code ---

```
Rscript inst/scripts/design_from_bedpe_cli.R \  
  "ACGT..." pairs.bedpe 1 0 genome.fa 1
```

--- code ---

1. 'SEQUENCE'
2. 'BEDPE_PATH'
3. 'SEQUENCE_GENOMIC_START'1
4. 'REGION_PADDING'0
5. 'GENOME_FASTA'
6. 'MAX_HITS'1

7.2 CLI

--- code ---

```
Rscript inst/scripts/design_taqman_cli.R \
  "ACGT..." 100 400 320 test_regions.tsv genome.fa 1
```

--- code ---

8.

- 'region_id'
- 'anchor_primer_seq', 'anchor_primer_start', 'anchor_primer_end', 'anchor_primer_tm', 'anchor_primer_gc'
- 'test_primer_seq', 'test_primer_start', 'test_primer_end', 'test_primer_tm', 'test_primer_gc'
- 'probe_seq_antisense', 'probe_start', 'probe_end', 'probe_tm', 'probe_gc'
- 'probe_gap_from_anchor_primer'
- 'pseudo_amplicon_size'
- 'score'

9.

9.1 'Sequence includes invalid bases'

A/C/G/T

9.2 'BEDPE column 7 anchor_flag must be A or B'

A B

9.3 '...outside sequence bounds'

BEDPE 'sequence'

- 'sequence_genomic_start'
- BEDPE
- 'region_padding'

9.4 0

- Tm/GC/distance/uniqueness
- anchor/test cut-site
- 5'GC>Ggap

- 'distance_to_cut'
- 'primer_tm_range'
- 'uniqueness_max_hits'
- 'amplicon_size_max'

10.

1. BEDPE
- 2.
3. top
4. /
- 5.

11.

- v0.4.0 BEDPEA/B Anchor +